Supplementary Material

Deep Neural Network Surrogates for Approximate Bayesian Computation:
A Simulation Study for Mutation-Rate Estimation
in Markov Branching Processes

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This document contains the full architecture-, activation-, and capacity-search tables referenced from the main text as "Supplementary Material S1", together with the full per-family results of Study II's architecture comparison and the validation of the fast simulator used in Study II, Part B (Section S4). Each table below corresponds to a numbered claim in the main text; the surrounding prose there states the headline conclusion, and is not repeated here except as needed to make each table self-contained.

S1 Study I: the full activation, normalisation and capacity search

The deployed one-dimensional surrogate (Section 4.1 of the main text) is a shallow funnel multilayer perceptron, input $\theta = \log_{10} p$, two GELU-activated dense layers at width 32–16 (626 parameters). That choice is the outcome of the two-stage search reported in full here: activation/normalisation first, at a 128–64 baseline, then capacity, once the activation question was settled.

Table S1: Architecture search, one-dimensional model: mean-curve fit MSE against the denoised response curve (irreducible replicate-noise floor $\approx 4.7 \times 10^{-3}$).

Model	Mean-curve MSE	vs. GP
Gaussian process (GPS-ABC baseline)	3.836×10^{-4}	—
DNN, GELU (128, 64), no BatchNorm [selects activation]	$\mathbf{3.888 \times 10^{-4}}$	+1%
DNN ensemble ×5, SiLU (128, 128, 64)	3.960×10^{-4}	+3%
DNN ensemble ×5, tanh (256, 128)	4.178×10^{-4}	+9%
DNN, SiLU (128, 128, 64)	4.400×10^{-4}	+15%
DNN, tanh (64, 64, 32)	4.435×10^{-4}	+16%
DNN, tanh (256, 128)	4.503×10^{-4}	+17%
DNN, ReLU + BatchNorm + dropout (64, 64, 32)	4.352×10^{-3}	+1035%

The row that determines every downstream comparison in the main text is the last one: an otherwise-reasonable network carrying BatchNorm and dropout fits *eleven times worse* than the same network without them. BatchNorm normalises per-minibatch statistics, useful in deep classification networks; but on a smooth scalar regression over a one-sided domain (the θ-grid has a hard edge at -8) it injects batch-dependent noise and biases predictions near that boundary. A follow-up ablation isolating only the activation function (same 128–64 architecture, no BatchNorm, averaged over three random seeds) found ReLU, SiLU, GELU, and tanh within 7% of one another

(3.95–4.23×10^{-4}): a statistical tie, with ReLU marginally ahead. Because accuracy does not separate them, the choice of GELU is made on structural grounds (smoothness, immunity to the dead-neuron failure mode) at no measurable accuracy cost. A second search round using 7-network deep ensembles (mean-curve MSE 3.94–4.25×10^{-4} across four ensemble configurations) did not improve on the single chosen network, so the simpler model is retained throughout.

Table S2: One-dimensional model: does capacity matter? Curve MSE is against the denoised response curve, mean over three seeds. Downstream MSE and 95% CI length are DNN-ABC at the same scale as the main text's mean-squared-error table (40 replicates/cell, full $p \times J$ grid), one seed per row.

Hidden layers	Parameters	Curve MSE ($\times$ GP)	Downstream MSE	Downstream 95% CI
GP (GPS-ABC baseline)	—	3.836×10^{-4} (—)	2.702×10^{-6}	2.137×10^{-3}
32–16 (deployed)	**626**	$\mathbf{3.896 \times 10^{-4}}$ (**+1.6%**)	$\mathbf{1.433 \times 10^{-6}}$	$\mathbf{1.300 \times 10^{-3}}$
128–64 (original)	8,642	4.129×10^{-4} (+7.6%)	1.514×10^{-6}	1.308×10^{-3}
64–32	2,274	3.957×10^{-4} (+3.2%)	1.565×10^{-6}	1.377×10^{-3}
16 (1 layer)	66	4.460×10^{-4} (+16.3%)	1.579×10^{-6}	1.375×10^{-3}
8 (1 layer)	34	4.983×10^{-4} (+29.9%)	1.668×10^{-6}	1.393×10^{-3}
16–8	186	4.508×10^{-4} (+17.5%)	1.810×10^{-6}	1.528×10^{-3}
Linear (control)	4	4.672×10^{-4} (+21.8%)	—	—

On raw curve fit, capacity is not merely unnecessary above a small threshold: it is mildly *counter-productive* at the original size: 128–64 has the largest gap to the GP of any two-layer candidate tested (+7.6%), while 32–16, at 14× fewer parameters, comes closest to matching it (+1.6%). On the downstream task the surrogate is actually used for, 32–16 is also the best point estimate on both accuracy and interval length of every candidate tested, 128–64 included, though attaching Monte Carlo standard errors to the gap shows this specific edge is not itself resolved (all nine Δ/SE values fall below 0.6 in magnitude). Nothing in the data argues against the smaller network either: 32–16 never does measurably worse than 128–64 in any cell. Only the linear control shows a clear, unambiguous cost, confirming some nonlinearity is required, just not much capacity beyond that.

S2 Study II: the full activation search

The deployed three-dimensional surrogate uses GELU into tanh (different activations per hidden layer) at width $64 \to 32$. Ten activations spanning three families (piecewise-linear: ReLU, LeakyReLU, PReLU; smooth saturating: tanh, ELU, SELU; and smooth unbounded: softplus, GELU, SiLU, Mish) were compared on the deployed network, five seeds each.

Two results follow, pulling in opposite directions. First, the choice *can* matter: the best-to-worst spread is 29% of the best row's MSE at $t = 2.7$, outside seed noise, and softplus fails outright (strictly positive, so every downstream layer must undo an accumulated offset). Second, the five leading candidates (GELU, tanh, LeakyReLU, ReLU and Mish) are mutually indistinguishable, every pairwise $|t|$ among them below 2. SELU's poor placement reflects that it presupposes LeCun-normal initialisation and AlphaDropout, absent under this project's standard initialisation; its position here is evidence about this configuration, not a general verdict on SELU.

Because the two hidden layers need not share an activation, all $10 \times 10 = 100$ ordered per-layer assignments were also compared. All 100 pairs are screened; the ten leading candidates are then re-fit on fifteen *fresh* seeds that played no part in selection, with the test split touched exactly once,

Table S3: Study II: activation comparison on the deployed $64 \to 32$ network, five seeds each. MSE is against held-out design-point means of $\log_{10} S$; the irreducible floor is 3.451×10^{-4}. The final column compares each row to the best, in units of the standard error of the difference. Standard deviations are across seeds (sample, $n - 1$).

Activation	Family	$\times$ floor	$\pm$ sd (seeds)	t vs. best
GELU	smooth unbounded	**1.072**	9.5×10^{-6}	—
tanh	smooth saturating	1.078	6.0×10^{-6}	0.42
LeakyReLU	piecewise-linear	1.080	6.4×10^{-6}	0.50
ReLU	piecewise-linear	1.082	9.5×10^{-6}	0.54
Mish	smooth unbounded	1.094	7.2×10^{-6}	1.38
SiLU	smooth unbounded	1.130	1.2×10^{-5}	3.00
PReLU	piecewise-linear	1.135	1.3×10^{-5}	3.04
ELU	smooth saturating	1.166	2.3×10^{-5}	2.95
SELU	smooth saturating	1.265	3.3×10^{-5}	4.35
softplus	smooth unbounded	1.383	8.7×10^{-5}	2.74

for the eventual winner alone.

Table S4: Study II: activation selection with no test-set leakage. All 100 ordered pairs were screened on validation; the ten finalists were re-fit on fifteen fresh validation seeds. Selection is on validation MSE; the test column is shown for reference and played no part in the choice. Test $\times$ floor is against the irreducible test floor 3.451×10^{-4}.

Layer 1 (64)	Layer 2 (32)	Val MSE	$\pm$ sd	t vs. best	Test $\times$ floor
GELU	**tanh**	$\mathbf{2.980 \times 10^{-4}}$	1.3×10^{-5}	—	1.062
Mish	tanh	3.006×10^{-4}	1.4×10^{-5}	0.54	1.076
SiLU	tanh	3.021×10^{-4}	1.4×10^{-5}	0.86	1.086
LeakyReLU	LeakyReLU	3.059×10^{-4}	8.4×10^{-6}	2.00	1.075
PReLU	ReLU	3.071×10^{-4}	1.2×10^{-5}	2.05	1.077
Mish	Mish	3.120×10^{-4}	1.1×10^{-5}	3.19	1.080
ELU	GELU	3.137×10^{-4}	8.5×10^{-6}	3.99	1.093
SiLU	Mish	3.164×10^{-4}	1.4×10^{-5}	3.84	1.097
GELU	Mish	3.172×10^{-4}	1.7×10^{-5}	3.51	1.086
Mish	SiLU	3.205×10^{-4}	1.8×10^{-5}	3.94	1.101

The selection optimism this two-stage discipline guards against is not small: the finalists perform, on average, 1.03×10^{-5} *worse* on the fresh seeds than on the seeds that chose them, 45% of the entire spread of the confirmation table and comparable to the across-seed standard deviation itself. The pair that screened best, SiLU into tanh, finished third on confirmation: a single-stage search would have reported that movement as a real result. Under the fourth-root target used throughout the main text, GELU into tanh is best on both validation and test, so scoring selection on test would not, here, have changed the choice; under the square-root target tried during development it did, which is why the two-stage, validation-only discipline is treated as a requirement rather than a formality.

S3 Study II: the architecture-family comparison

A 1-D CNN (two convolutional layers, kernel width 3), a GRU, and an LSTM (each unrolling the three inputs $(\log_{10} p_1, \log_{10} p_2, \tau)$ as three timesteps), with comparable parameter counts (2,402–3,426) to the deployed FFN, were each evaluated exactly as GPS-ABC and DNN-ABC were: a surrogate-fit comparison against the irreducible noise floor, and the identical full ABC-MCMC recovery task (same three truth triples, 16 replicates, 3,000 iterations, 1,000 burn-in).

Table S5: Surrogate fit quality by architecture family, test set. The irreducible noise floor is 3.451×10^{-4}. The deployed FFN's row is read from the capacity study rather than restated, so it necessarily describes the same target as the rows beside it.

Architecture	Parameters	MSE (design means)	× floor	R^2	95% coverage
CNN1D (2 conv layers, k=3)	2,482	3.589×10^{-4}	1.04	0.99670	0.949
FFN 64–32 [deployed]	**2,402**	$\mathbf{3.749 \times 10^{-4}}$	**1.09**	**0.99656**	**0.952**
LSTM (hidden 24)	2,642	4.221×10^{-4}	1.22	0.99612	0.950
RNN (GRU, hidden 32)	3,426	4.520×10^{-4}	1.31	0.99585	0.949

Table S6: ABC-MCMC recovery of (p_1, p_2, τ) by architecture family, per-parameter RMSE ($\log_{10}$ scale for p_1, p_2; absolute time units for τ), averaged over the three truth triples used throughout Study II. Bold marks the best value in each row.

Parameter	GPS-ABC	DNN-ABC (FFN)	CNN1D-ABC	RNN-ABC	LSTM-ABC
p_1 RMSE	1.391	1.407	**1.373**	1.378	1.389
p_1 coverage	1.000	0.938	0.958	0.917	0.979
p_2 RMSE	**0.156**	0.258	0.203	0.312	0.209
p_2 coverage	1.000	0.979	0.979	0.938	0.958
τ RMSE	**1.422**	1.753	1.688	1.700	1.631
τ coverage	1.000	1.000	1.000	1.000	1.000

CNN1D statistically ties the deployed FFN (4.3% lower MSE, inside the seed-to-seed spread already documented for the capacity study), but costs 143 µs per query against the FFN's 40, the wrong trade for a tie. LSTM and RNN are measurably worse (+12.6%, +20.6% MSE; both gaps an order of magnitude larger than their own across-seed standard deviation). Attaching Monte Carlo standard errors to every one of the $3 \times 3 \times 3 = 27$ new-family, truth and parameter cells, and comparing each new family against both GPS-ABC and DNN-ABC at the 2-MCSE level (54 pairwise comparisons in total), gives a picture consistent with the fit-quality result, and a slightly starker one: 49 of 54 (91%) are statistical ties. All 5 resolved comparisons favour GPS-ABC, and *none* favours a new family anywhere. Three of the five are CNN1D-ABC and two RNN-ABC; LSTM-ABC is tied in every one of its eighteen comparisons.

Interval calibration adds nothing to the picture either way. Pooled coverage is 0.951–0.979 for the four neural families and 1.000 for both Gaussian processes, against a nominal 0.95. An earlier version of this comparison read a narrative into that column (the FFN under-covering, CNN1D inheriting and worsening it, LSTM correcting it) which the corrected numbers do not support: the spread across all four neural families is under three percentage points, which at sixteen replicates is at most one cell per parameter. Every method over-covers on τ at exactly 1.000, reflecting that parameter's weak identifiability rather than any property of an architecture.

S4 Study II, Part B: validation of the fast simulator

The corrected fast simulator of Section 5.10.1 of the main text is compared with the exact cell-by-cell simulator at $t_p = 10$ and $t_p = 12$, the only scales at which the latter is feasible. For each setting the same number of cultures is drawn from each simulator and the per-culture statistic $(X/Z)^{1/4}$ is compared with a two-sample Kolmogorov–Smirnov test. Rates are chosen so that the expected number of mutation events per culture spans the same range as the published regime. The smallest p-value over the twelve settings is 0.015, above the Bonferroni threshold of 0.004; the largest KS distance is 0.020. The largest relative gap in the mean statistic is 5.8% (at $p_1 = 10^{-4}$, $p_2 = 10^{-5}$, $\tau = 6$), which the KS test does not reject ($p = 0.20$).

Table S7: Fast versus exact two-stage simulator: mean per-culture statistic $(X/Z)^{1/4}$ and Kolmogorov–Smirnov comparison. n cultures per simulator.

p_1	p_2	τ	t_p	n	exact	fast	KS D	KS p
10^{-5}	10^{-5}	5	10	12,000	0.0337	0.0335	0.0098	0.615
10^{-4}	10^{-4}	5	10	12,000	0.1395	0.1396	0.0063	0.968
10^{-5}	10^{-4}	3	10	12,000	0.1386	0.1383	0.0092	0.691
10^{-4}	10^{-5}	6	10	12,000	0.0431	0.0411	0.0138	0.205
10^{-3}	10^{-5}	2	10	12,000	0.0377	0.0374	0.0103	0.550
3×10^{-4}	2×10^{-5}	7	10	12,000	0.1029	0.1008	0.0103	0.550
3×10^{-4}	2×10^{-5}	9	10	12,000	0.1677	0.1650	0.0202	0.015
10^{-2}	10^{-4}	1	10	12,000	0.1460	0.1456	0.0082	0.815
2×10^{-3}	8×10^{-3}	5	10	12,000	0.5447	0.5450	0.0176	0.048
10^{-6}	10^{-6}	6	12	3,000	0.0161	0.0161	0.0083	1.000
10^{-5}	10^{-6}	4	12	3,000	0.0163	0.0172	0.0140	0.930
10^{-6}	10^{-5}	8	12	3,000	0.0706	0.0700	0.0140	0.930